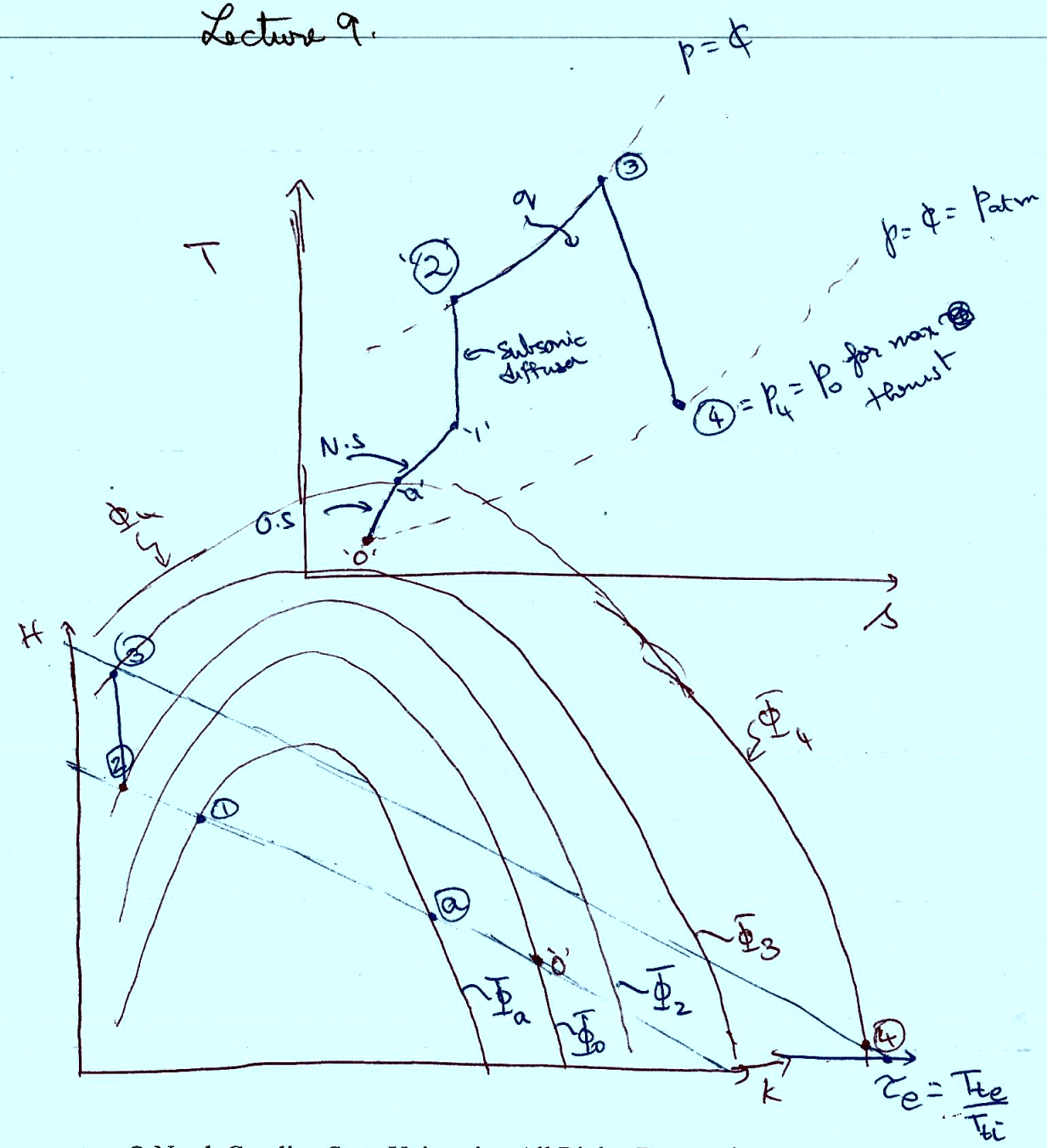
2/12/19

Lecture 9.

In T-S space



In H-K space

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$0 \to a$ O.S. decel. of supersonic flow, effective area decreases, $\Phi \downarrow$ $\Phi_a < \Phi_0$

$a \to 1$ N.S. $\Phi = \phi$; $\Phi_1 = \Phi_a$

$1 \to 2$ Subsonic decel. $\Phi \uparrow$.

$2 \to 3$ Isobaric heat add $\Rightarrow \Phi \uparrow \longrightarrow H + k = \tau_e$; $k = \phi$;

$3 \to 4$ accel. from subsonic to supersonic (adiabatic)

$$M_c \leq M_0$$

$$\Phi_a = \Phi_1 < \Phi_0 < \Phi_2 < \Phi_3 < \Phi_4$$

$$H_4 > H_0$$

$$k_4 > k_{\phi 0}$$

$$\frac{\tau}{\dot{m}(C_p T_{ti})^{1/2}} = \Phi_4 - \Phi_0 > 0 \Rightarrow \text{thrust}$$

- Just based on h-k diagram, it is <u>not</u> required to have an O.S. to go from ⓪ to 'a'. We could go from 'a' to '①' with a single normal shock & this pt ① would be @ a higher Φ compared to with O.S. But the h-k diagram does not show the entropy generation (or loss in p_t). In reality, we want the distance between ⓐ & ① to be as short as possible (We want to cause the vel. change through O.S. & area change & not using N.S.)

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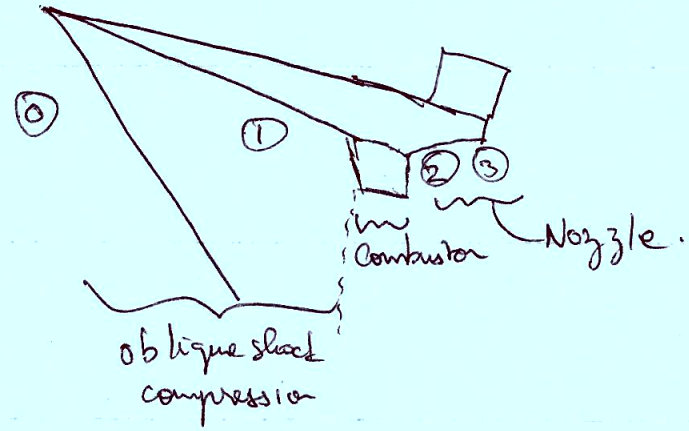
Scramjets:



→ Flow is decelerated from $M \geq 6$ to $M_{comb} \approx 1.3 - 2.3$ thru one or more oblique shocks.

→ Must avoid normal shocks.

→ O.S. introduces an effective area change.

$0 \rightarrow 1 \qquad \Phi \downarrow$

$H + k = 1$

→ In combustor, hydrocarbon (C_xH_y) or hydrogen is injected & burned forming combustion products. Again we have 2 option on how to burn the fuels:

a) const area : $\Phi = \phi$; $k \downarrow$ (supersonic comb.)

b) const pr : $\Phi \uparrow$; $k = \phi$

During heat add^n, we move from $H + k = 1$ to $H + k = \tau_e$

Regardless of option (a) or (b), both will decrease Mach #.

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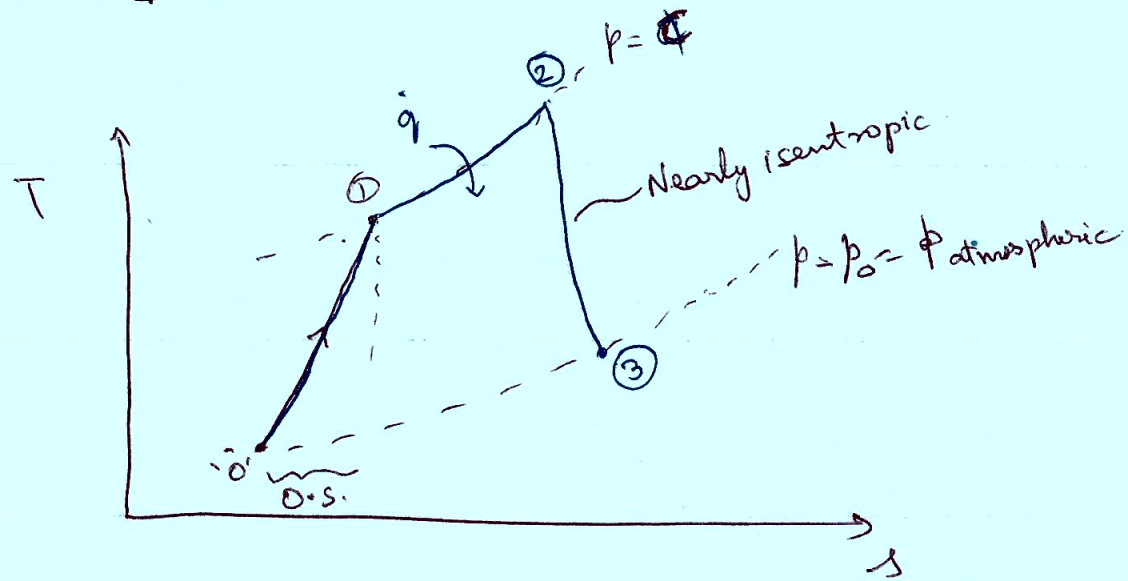
→ Must be careful not to introduce N.S. due to excessive drag losses; but also need to penetrate the fuel into air stream & get it to mix as quickly as possible

→ For combustion to occur, need to mix O_2 & C_xH_y @ molecular level → big challenge!!

→ Flow is accelerated from supersonic ($M \approx 2$) to hypersonic in the nozzle. Just as in ideal ramjet, for ideal scramjet $M_e = M_0$; for non-ideal $M_e \leq M_0$

② → ③ $\Phi \uparrow$ adiabatic isocontour $H + k = T_e$

In T-s space.



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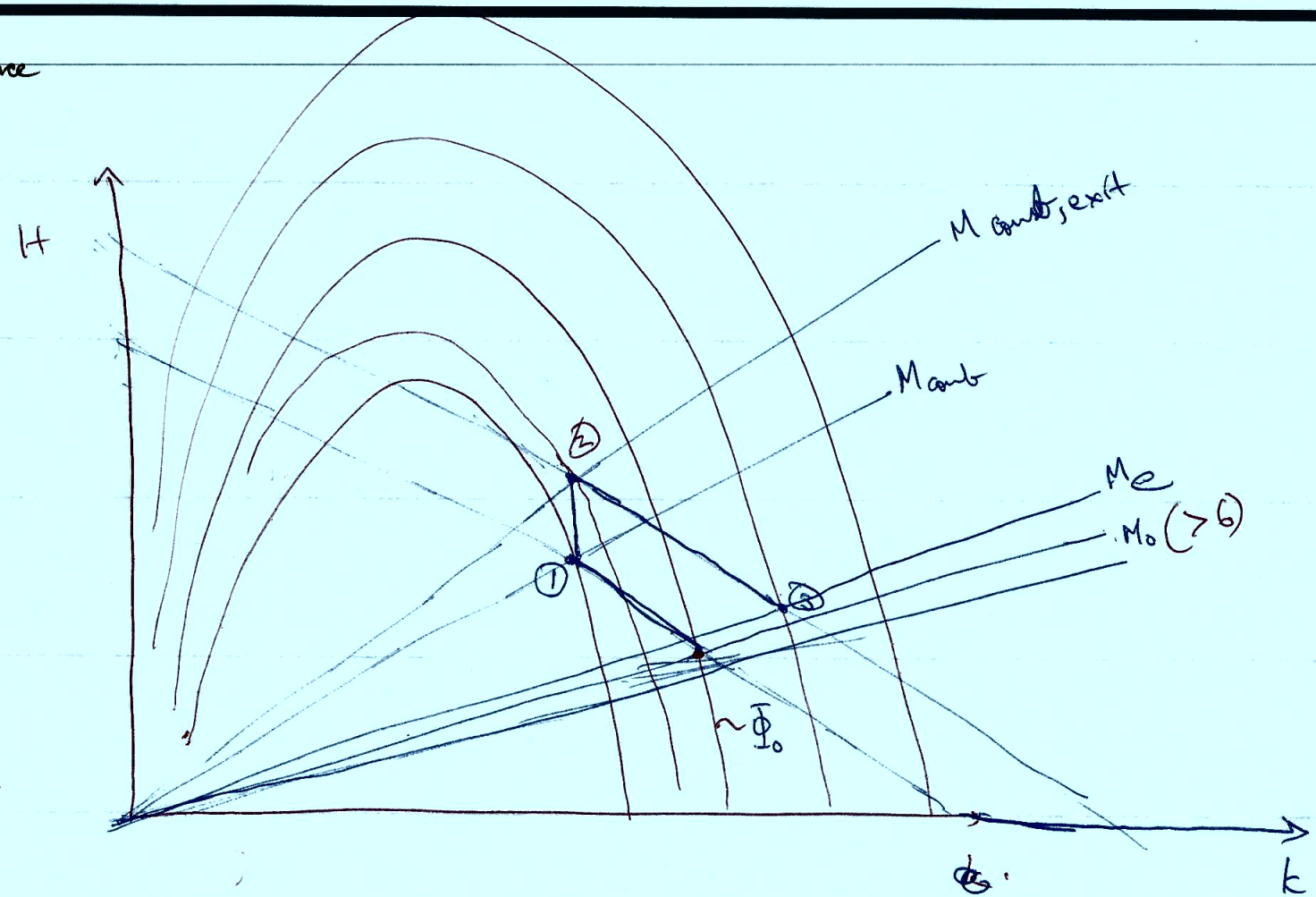
In h–k space



HW2 : Ch2 : 17, 18 (a&b), 19, 21, 22.

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Another note on H & k

$$\frac{k}{H} = \frac{V^2}{2C_pT_{ti}} \cdot \frac{C_pT_{ti}}{C_pT} = \frac{V^2}{2C_pT} = \frac{\gamma R T M^2}{2C_pT}$$

$$\frac{k}{H} = \frac{\gamma R}{C_p} M^2$$

$$\boxed{\frac{k}{H} = \frac{\gamma - 1}{2} M^2} =$$

$$\frac{k}{H} = \frac{\gamma - 1}{2} M^2 = \frac{\text{kinetic En}}{\text{Static enthalpy}}$$

- @ Low Mach #, most of the total temp, comes from static enthalpy, & Mach # changes are due primarily to changes in vel.
- In transonic regime ($M \approx 1$), both static enthalpy & K.E are important in determining total temp. Mach # changes are due to both changes in vel. & speed of sound.
- In supersonic regimes ($1 \leq M \leq 6$), K.E begins to dominate
- In hypersonic regimes ($M > 6$), KE can be $\underbrace{O(100)}_{\text{order of } 100} \times$ static enthalpy.

Mach # changes are due primarily to changes in the speed of sound.

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Know that dissociation becomes important @ about 1700K. What is the Mach # corresponding to this stag temp?

$$\frac{T_t}{T_0} = 1 + \frac{\gamma - 1}{2} M^2$$

$$T_0 = 220K, \ \gamma = 1.4.$$

$$\Rightarrow \underline{\underline{M = 6}}$$

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